

Efficient ICI and Off-Grid Effect Suppression Algorithm for ISAC-Enabled OFDM Radar

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Abstract—Orthogonal frequency-division multiplexing (OFDM) has drawn a lot of attention in the study of integrated sensing and communications (ISAC). However, the fundamental drawback of Doppler intolerance limits its applicability in highly dynamic scenarios such as vehicular applications. In addition, the delay and Doppler shift that are non-integer times the resolution cause the off-grid effect, leading to scalloping loss and increasing the sidelobes. In this work, a super-resolution range-doppler map (RDM) modeling algorithm with low complexity is proposed to suppress these impacts. The proposed algorithm models the RDM contributed by the detected targets jointly via 2D surface fitting and then reconstructs and subtracts the corresponding signal components from the original received signal. Consequently, the weaker targets hiding under the inter-carrier interference (ICI) and off-grid effect become detectable. The communications capability is not impacted since the algorithm is only deployed at the radar receiver. In addition, two variants of the algorithm are illustrated, showing a trade-off between accuracy and complexity. Simulation results demonstrate that the proposed algorithms can effectively suppress ICI and mitigate off-grid effects, while simultaneously enhancing the accuracy of delay and Doppler estimation. It is indicated that the proposed algorithms can provide sufficient improvements in multi-target scenarios.

Index Terms— 6G, automotive radar, Doppler effect, integrated sensing and communications (ISAC), mmWave, orthogonal frequency-division multiplexing (OFDM), vehicle-to-everything (V2X).

I. INTRODUCTION

Automotive radar enables vehicles to perceive their surroundings, providing an improved safety and driving experience. The millimeter-wave (mmWave) band (30 GHz–300 GHz) automotive radar has become the focus of research recently due to its relatively large bandwidth, which enables higher resolution [1]. Advanced automotive radar systems primarily operate in the 77 GHz band, with the 76–77 GHz band allocated for long-range radar and the 77–81 GHz band designated for short- and mid-range applications [2], [3]. Meanwhile, vehicular information systems based on vehicle-to-everything (V2X) communications allow vehicles to share information such as their status, road conditions, and traffic management information, enhancing transportation efficiency and safety [4].

This work was supported by the Federal Ministry of Education and Research (BMBF) of Germany partly in the project ‘KOMSENS-6G’ (grant number: 16KISK126) and ‘Open6GHub’ (grant number: 16KISK005). (*Corresponding author: Yanpeng Su.*)

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Automotive radar and vehicular communications are regarded as two key technologies enabling autonomous driving in the future [4], [5]. To overcome the potential spectrum scarcity and interference between two systems, the idea of integrating the two functionalities into one system has been widely discussed in recent years, known as integrated sensing and communications (ISAC) [6], joint communications and sensing (JCAS) [7], or joint radar and communications (JRC) [8]. Among the existing waveforms, orthogonal frequency-division multiplexing (OFDM) is an attractive candidate for ISAC and has been widely studied. Compared with radar-centric waveforms such as frequency-modulated continuous waveform (FMCW) and phase-modulated continuous waveform (PMCW), OFDM allows extremely high throughput transmission in multipath fading channels and has been applied in communications networks from 4G long term evolution (LTE) to 5G new radio (NR), while OFDM radar technology has been widely studied and is gradually maturing [9]–[12]. However, OFDM radar has an obvious drawback of Doppler intolerance, as the orthogonality between the subcarriers can be destroyed by the Doppler effect, leading to inter-carrier interference (ICI) and potentially threatening its performance in highly dynamic scenarios such as vehicular applications.

Methods for suppressing ICI of OFDM ISAC systems are addressed in [10], [13]–[20]. In [10], an all-cell Doppler correction (ACDC) method correcting the phase shift produced by the Doppler effect before OFDM demodulation is proposed, which can effectively suppress ICI regardless of the number of reflection paths. However, the ACDC method is only compatible with repeated symbols OFDM (RS-OFDM) [13], where the OFDM symbols in the slow time domain are the same, severely reducing communications throughput. Studies in [14], [15] can be regarded as an extension of ACDC, thus sharing the same problem in communications capability. Methods for estimating the Doppler shift based on correlation properties of the subcarrier symbols are introduced in [16]–[18]. However, these methods can address Doppler ambiguity and ICI issues only in the scenario of a single target that is assumed to be detectable. In addition, these methods are only applicable to phase-coded OFDM (PC-OFDM) systems since the performance highly relies on the autocorrelation property. To maintain a good autocorrelation function (ACF) on the subcarriers, compromise on data rate is inevitable. ICI suppression via beamforming design is discussed in [19], which is also under the assumption of a single detectable target. In [20], a signal reconstruction-based communications-compatible ICI mitigation method is proposed, where the reflection signals

are reconstructed separately and canceled from the received signal. The method extremely increases the complexity since range-Doppler map (RDM) estimation needs to be performed for each target separately.

However, the off-grid effect is not considered in the above-mentioned studies, which may lead to scalloping loss and also increase sidelobes on the RDM [21]. The off-grid effect comes from mismatched fast Fourier transform (FFT) or inverse fast Fourier transform (IFFT) when the delay and Doppler shift are non-integer times their resolution. The performance of target-dependent ICI suppression methods may be impacted by the off-grid effect. For instance, the bias in estimated delay and Doppler as well as the scalloping loss lead to an incompletely eliminated reflection signal as well as ICI with the algorithm in [20].

The off-grid effect can be simply suppressed by applying window functions, which extend the width of the mainlobe and suppress the sidelobes, while the ICI is not restrained in this process. Relevant works [21]–[27] mainly concentrate on fine FFT/IFFT estimation instead of sidelobe suppression, while the Doppler intolerance is not addressed. The conventional interpolation method is implemented in [22], while the methods of upsampling or zero-padding are investigated in [23]–[25]. Methods of fine estimation via linear approximation can be found in [26], [27], which involve the samples in the mainlobe to calculate the accurate Doppler shift. The interpolation by optimization method is discussed in [21], where the delay and Doppler are fine-estimated via a convex optimization problem. The optimization method can mitigate the off-grid effect with a high computational cost as the RDM estimation should be implemented repeatedly.

To suppress the impact of ICI and off-grid effect jointly with limited computational effort, in this work, a super-resolution RDM modeling algorithm is proposed, allowing for the suppression of ICI and the off-grid effect. The method involves the processes of RDM modeling, fine estimation, signal reconstruction, and cancellation. The RDM modeling is a target-dependent interpolation by optimization method that avoids repeating RDM calculations while considering the interference between target points on the RDM. While the proposed algorithm significantly suppresses the impact of the ICI and the off-grid effect, a super-resolution range and Doppler estimation is achieved simultaneously. The impact on communications capability is avoided as the algorithm is applied at the radar receiver. In addition, its compatibility with various waveforms is explained in this paper. Our contribution is summarized as follows:

- **ICI and off-grid effect suppression algorithm via RDM modeling and signal cancellation.** The delay, Doppler, and amplitude of the reflection signals are fine-estimated in the RDM modeling process and used to reconstruct the received signal components. The cancellation process removes the detected targets' contribution including ICI and sidelobes on the RDM, thus the weaker targets are observable. The proposed algorithm can provide a reliable performance in multi-target scenarios without impacting the communications capability.

- **Two variants of the suppression algorithm: trade-off between accuracy and complexity.** On the one hand, in the case that the window function is involved in RDM estimation, the effective suppression of the sidelobes allows independent processing of the targets, and the 2D RDM modeling problem can be divided into two 1D curve fitting problems for the range and Doppler profiles, simplifying the estimation process. On the other hand, considering the RDM modeling process is inevitably affected by ICI, the method proposed in [20] can be integrated to improve the proposed algorithm. However, while the RDM cell noise is more thoroughly eliminated, the complexity is significantly increased.
- **Use cases and versatility for other waveforms:** Based on the properties of the proposed algorithms, the use cases such as traffic and vehicular applications are analyzed. Additionally, the compatibility of the proposed algorithms with other waveforms like FMCW and PMCW is explained.

The rest of this paper is organized as follows: Section II reviews the OFDM radar system model and states the problem of ICI and the off-grid effect. The proposed algorithm as well as its two variants are introduced in Section III. The simulation configuration and results are presented in Section IV, where the performance of the proposed algorithms in different scenarios is investigated. The advantages and disadvantages of the proposed algorithms are summarized in Section V, then the use cases are analyzed. Section VI concludes this work.

Notations: $()^*$, $()^T$, $()^H$, and $()^{-1}$ denote the complex conjugate, transpose, conjugate transpose, and inverse, respectively. a and A represent scalars, $\mathbf{a}$ and $\mathbf{A}$ respectively represent vectors and matrices. The diagonalization calculation of vector $\mathbf{a}$ is written by $\text{diag}(\mathbf{a})$. $\mathbf{A} \odot \mathbf{B}$ and $\mathbf{A} \oslash \mathbf{B}$ imply Hadamard product and element-wise division of $\mathbf{A}$ and $\mathbf{B}$. $\mathbf{A}(n, m)$ implies the component at the n -th row and m -th column of matrix $\mathbf{A}$. $\mathbb{Z}$, $\mathbb{N}$, $\mathbb{R}$, and $\mathbb{C}$ express the sets of integers, natural numbers, real numbers, and complex numbers. $\mathcal{CN}(\mu, \sigma^2)$ denotes complex normal distribution with mean and variance of μ and σ^2 .

II. SYSTEM MODEL AND PROBLEM STATEMENT

The model of OFDM radar has been discussed in [21], [28], [29]. This section provides a review of its architecture and then estimates the RDM shape contributed by the targets. Afterward, the problems of ICI and the off-grid effect are illustrated.

A. OFDM Radar Model

Fig. 1 illustrates the OFDM radar signal processing chain. The transmitter continuously generates $N \times M$ modulated data symbols to form a radar frame, where N equals the number of subcarriers and M is the number of OFDM symbols. The generated baseband signal after OFDM modulation is given by

$$x(t) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} \mathbf{X}(n, m) e^{j2\pi n \Delta f (t - (mT_O + T_{CP}))} \cdot \text{rect}\left(\frac{t - (mT_O + T_{CP})}{T_O}\right), \quad (1)$$

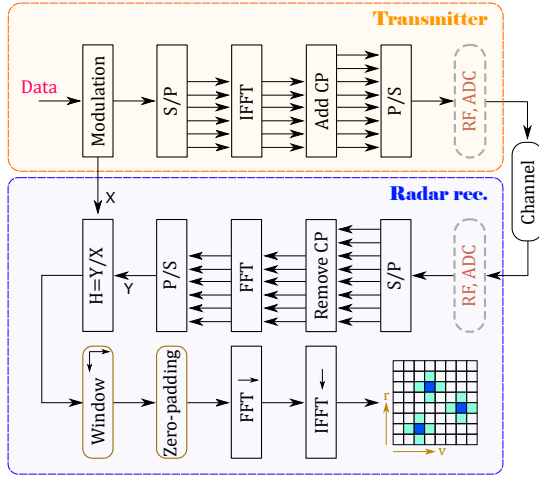


Fig. 1. OFDM radar architecture. The radar receiver processes the received signal using the periodogram-based algorithm. The radar receiver outputs a range-Doppler map.

where $\mathbf{X}(n, m)$ represents the modulated data symbol on the n -th subcarrier and m -th OFDM symbol. Δf is the subcarrier distance, T_{CP} indicates the cyclic prefix (CP) length. $T_O = T + T_{CP}$ is the total OFDM symbol duration including the cyclic prefix and the OFDM symbol length T , and $T = 1/\Delta f$.

After adding CP, the signal is shifted to the carrier frequency and transmitted. After interaction with the environment, the reflected and scattered signals are collected by the radar receiver and then moved back to the baseband

$$r(t) = \sum_{k=0}^{K-1} A_k x_{CP}(t - \tau_k) e^{j2\pi f_{D,k} t} + z(t), \quad (2)$$

where K represents the number of propagation paths (targets). $x_{CP}(t)$ denotes the CP-added transmitted signal. The complex amplitude (CA) $A_k = a_k e^{j(-2\pi f_c \tau_k + \varphi_k)}$ combines all time-irrelevant components including attenuation and constant phase shifts. a_k , τ_k , $f_{D,k}$, and φ_k denote the attenuation, delay, Doppler shift, and constant phase shift of the path k . f_c represents the carrier frequency. $z(t)$ implies the channel additive white Gaussian noise (AWGN).

After being processed by the analog-to-digital converter (ADC) and CP removal, assuming $\tau_k \leq T_{CP}, \forall k$, i.e., there is no inter-symbol interference (ISI), the resulting signal is formed as an $N \times M$ matrix $\mathbf{R}$ with element on the n -th row and m -th column

$$\mathbf{R}(n, m) = \sum_{k=0}^{K-1} \sum_{\mu=0}^{N-1} A_k \mathbf{X}(\mu, m) e^{j2\pi n \frac{\mu + \bar{f}_{D,k}}{N}} \cdot e^{-j2\pi \mu \bar{\tau}_k} e^{j2\pi \bar{f}_{D,k} m \alpha} + \mathbf{Z}(n, m), \quad (3)$$

where $\alpha = T_O/T$, $\bar{f}_{D,k} = f_{D,k}/\Delta f$ and $\bar{\tau}_k = \tau_k/T$ denote the normalized Doppler shift and delay. $e^{-j2\pi \mu \bar{\tau}_k}$ and $e^{j2\pi \bar{f}_{D,k} m \alpha}$ are the impact of delay and Doppler. $e^{j2\pi n \frac{\bar{f}_{D,k}}{N}}$ implies the shift in frequency domain. $\mathbf{Z}$ is the channel AWGN component. As a result, $\mathbf{R}$ can be written as

$$\mathbf{R} = \sum_{k=0}^{K-1} A_k \mathbf{D}_N(\frac{\bar{f}_{D,k}}{N}) \mathbf{F}_N^{-1} \mathbf{D}_N(-\bar{\tau}_k) \cdot \mathbf{X} \mathbf{D}_M(\bar{f}_{D,k} \alpha) + \mathbf{Z}, \quad (4)$$

where $\mathbf{F}_N = \left[e^{-j2\pi \frac{\mu\nu}{N}} \right]_{\substack{0 \leq \mu \leq N-1 \\ 0 \leq \nu \leq N-1}} \in \mathbb{C}^{N \times N}$ represents the discrete Fourier transform (DFT) matrix, and the inverse discrete Fourier transform (IDFT) matrix is defined as $\mathbf{F}_N^{-1}$. And $\mathbf{D}_N(x)$ is given by

$$\mathbf{D}_N(x) = \text{diag}\{1, e^{j2\pi x}, \dots, e^{j2\pi(N-1)x}\}, \quad (5)$$

After FFT, the finally received symbols are obtained:

$$\mathbf{Y} = \mathbf{F}_N \mathbf{R}. \quad (6)$$

$\mathbf{Y}$ and $\mathbf{X} \in \mathbb{C}^{N \times M}$ respectively denote the received and transmitted symbols. The component $\mathbf{D}_N(\frac{\bar{f}_{D,k}}{N})$ shows the ICI due to the Doppler effect. $\mathbf{D}_N(-\bar{\tau}_k)$ denotes the continuously changed phase shift along subcarriers caused by delay, and $\mathbf{D}_M(\bar{f}_{D,k} \alpha)$ represents the impact of the Doppler effect in the slow time domain.

The channel response $\mathbf{H}$ is estimated by $\mathbf{H} = \mathbf{Y} \oslash \mathbf{X}$, with elements

$$\begin{aligned} \mathbf{H}(n, m) = & \sum_{k=0}^{K-1} A_k e^{-j2\pi n \bar{\tau}_k} e^{j2\pi \bar{f}_{D,k} m \alpha} \underbrace{\left(\text{sinc}(-\bar{f}_{D,k}) \right)}_{\mathbf{H}_{\text{des}}(n, m)} \\ & + \underbrace{\sum_{\mu=0, \mu \neq n}^{N-1} \frac{\mathbf{X}(\mu, m)}{\mathbf{X}(n, m)} \text{sinc}(n - \mu - \bar{f}_{D,k})}_{\mathbf{H}_{\text{ICI}}(n, m)} + \mathbf{H}_Z(n, m), \end{aligned} \quad (7)$$

where $\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$, $\mathbf{H}_{\text{des}}$ represent the desired component, $\mathbf{H}_{\text{ICI}}$ and $\mathbf{H}_Z$ denote the noise parts of ICI and channel AWGN. The ICI is caused by the Doppler effect as it leads to a shift in the frequency domain, increasing the noise on the RDM. For compatibility with the communications function, the OFDM symbols are generally randomly generated, thus the ICI approximately leads to AWGN on the RDM [20]. Therefore, $\mathbf{H}_{\text{ICI}}$ and $\mathbf{H}_Z$ are combined as $\mathbf{H}_{\text{ICI,Z}}$ in the following.

$\mathbf{H}_{\text{des}}$ is the desired component and can be written as

$$\mathbf{H}_{\text{des}} = \sum_{k=0}^{K-1} A_k b_k \mathbf{D}_N(-\bar{\tau}_k) \mathbf{1}^{N \times M} \mathbf{D}_M(\bar{f}_{D,k} \alpha), \quad (8)$$

where $b_k = \text{sinc}(-\bar{f}_{D,k})$ represents the loss due to the Doppler effect. $\mathbf{1}^{N \times M}$ denotes the all-ones matrix with size of $N \times M$.

The windowing and zero-padding processes help suppress the sidelobes and improve sensing accuracy. The windowing process is an element-wise multiplication of the input matrix and the window matrix, i.e., $\mathbf{H}_w = \mathbf{H} \odot \mathbf{U}$, where $\mathbf{U} = \mathbf{u}_d \otimes \mathbf{u}_v^T \in \mathbb{C}^{N \times M}$ expresses the 2D window matrix which is the outer product of the windows $\mathbf{u}_d \in \mathbb{C}^{N \times 1}$ and $\mathbf{u}_v^T \in \mathbb{C}^{1 \times M}$ applied in range (columns) and velocity (rows) profiles, respectively. Their Fourier transforms are denoted by $\mathbf{W}$, $\mathbf{w}_d$, and $\mathbf{w}_v$, respectively.

The zero-padding process directly adds zeros at the end of the columns and rows, i.e.,

$$\mathbf{H}_p = \begin{bmatrix} \mathbf{H}_w & \mathbf{0}^{N \times p_v M} \\ \mathbf{0}^{p_d N \times M} & \mathbf{0}^{p_d N \times p_v M} \end{bmatrix} \in \mathbb{C}^{N_p \times M_p}, \quad (9)$$

where $p_d = (N_p - N)/N$ and $p_v = (M_p - M)/M$ are the padding factors in the columns and rows, respectively.

After being processed by the row-wise FFT and column-wise IFFT, the finally obtained RDM matrix is given by

$$\mathbf{G} = \mathbf{F}_{N_p}^{-1} \mathbf{H}_p \mathbf{F}_{M_p}. \quad (10)$$

The element at the n -th row and m -th column of $\mathbf{G}$ is given by

$$\begin{aligned} \mathbf{G}(n, m) &= \sum_{\mu=0}^{N-1} \sum_{\nu=0}^{M-1} \mathbf{H}(\mu, \nu) \mathbf{W}(\mu, \nu) e^{-j2\pi \frac{m\nu}{M_p}} e^{j2\pi \frac{n\mu}{N_p}} \\ &= \mathbf{G}_{\text{des}}(n, m) + \mathbf{G}_{\text{ICLZ}}(n, m), \end{aligned} \quad (11)$$

where $\mathbf{G}_{\text{ICLZ}}$ is the AWGN component contributed by $\mathbf{H}_{\text{ICLZ}}$. The desired component contributed by $\mathbf{H}_{\text{des}}$ is given by

$$\begin{aligned} \mathbf{G}_{\text{des}}(n, m) &= \sum_{k=0}^{K-1} A_k b_k \sum_{\mu=0}^{N-1} \mathbf{D}_N(-\bar{\tau}_k) \mathbf{w}_d(\mu) e^{j2\pi \frac{n\mu}{N_p}} \\ &\quad \cdot \sum_{\nu=0}^{M-1} \mathbf{w}_v^T(\nu) \mathbf{D}_M(\bar{f}_{D,k} \alpha) e^{-j2\pi \frac{m\nu}{M_p}} \\ &= \sum_{k=0}^{K-1} A_k b_k \mathbf{W}\left(\frac{n - \bar{\tau}_k N_p}{1 + p_d}, \frac{m - \bar{f}_{D,k} \alpha M_p}{1 + p_v}\right). \end{aligned} \quad (12)$$

As a result, while the part of $\mathbf{G}_{\text{ICLZ}}$ results in AWGN on the RDM, the desired part $\mathbf{G}_{\text{des}}$ is equivalent to the superposition of the window functions $\mathbf{W}$ with different amplitudes and shifted positions corresponding to the CA, Doppler shift, and delay of the reflection signals.

B. Impact of ICI and Off-Grid Effect

The OFDM radar estimation process brings a processing gain of NM/SNR_w , where SNR_w denotes the signal-to-noise ratio (SNR) loss due to the window function. The power of desired target contribution is amplified by $(NM)^2/\text{SNR}_w$ times, the distribution of AWGN is changed from $\mathbf{H}_{\text{ICLZ}} \sim \mathcal{CN}(0, \sigma_{\text{ICI}}^2 + \sigma_{\text{Z}}^2)$ to $\mathbf{G}_{\text{ICLZ}} \sim \mathcal{CN}(0, NM(\sigma_{\text{ICI}}^2 + \sigma_{\text{Z}}^2))$ with a power gain of NM .

While the ICI increases the AWGN on the RDM, the off-grid effect is included in the desired part and mainly increases the sidelobes in the range or velocity profile. The delay and Doppler shift can be divided into an on-grid element and an off-grid element, i.e.,

$$\begin{aligned} \tau &= (\delta + \varepsilon) \Delta \tau_p = \frac{(\delta + \varepsilon)T}{N_p}, \\ f_D &= (\rho + \epsilon) \Delta f_{D,p} = \frac{(\rho + \epsilon)}{M_p T_O}, \\ \delta &\in \mathbb{N}, \quad \rho \in \mathbb{Z}, \quad \varepsilon, \epsilon \in (-0.5, 0.5], \end{aligned} \quad (13)$$

where $\Delta \tau_p = \frac{\Delta \tau}{1 + p_d}$ and $\Delta f_{D,p} = \frac{\Delta f_D}{1 + p_v}$ are the delay and Doppler resolutions with zero-padding, while the numerators are the resolutions without padding. δ and ρ represent the on-grid factors and determine the position of the peaks on the RDM. ε and ϵ denote the off-grid factors. $(\delta + \varepsilon, \rho + \epsilon)$ is the realistic target coordinate on the RDM.

Rewriting (12) with the consideration of the off-grid factors:

$$\mathbf{G}_{\text{des}}(n, m) = \sum_{k=0}^{K-1} A_k b_k \mathbf{W}\left(\frac{n - (\delta_k + \varepsilon_k)}{1 + p_d}, \frac{m - (\rho_k + \epsilon_k)}{1 + p_v}\right). \quad (14)$$

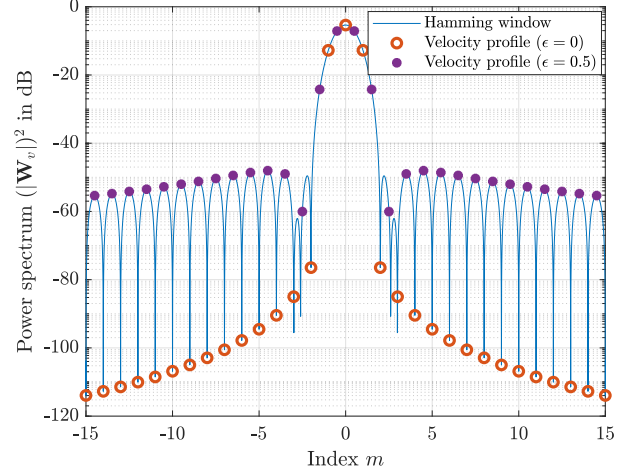


Fig. 2. Velocity profile with on-grid and off-grid Doppler effects. The blue line expresses the power spectrum of the Hamming window with a length of 1024. The power is normalized and the mainlobes of the curves are centered at $m = 0$.

The non-zero off-grid factors ε and ϵ lead to mismatched FFT and IFFT, increasing the sidelobes in the range and velocity profiles, respectively. For instance, Fig. 2 shows the power spectrum of normalized velocity profile with $\epsilon = 0$ (red hollow circles) and 0.5 (purple solid circles). The x -axis shows the relative horizontal index to the center of the mainlobe. The blue curve represents the power spectrum of the Hamming window as a reference. For better visibility, the diagram shows the thirty samples centered on the mainlobe. While the sidelobes are located at the troughs in the case of $\epsilon = 0$, the off-grid effect significantly increases the sidelobe level and slightly reduces the mainlobe power. For the systems without the window function (rectangular window), the off-grid effect can reduce the peak-to-sidelobe ratio (PSLR) even to $|\text{sinc}(0.5)/\text{sinc}(1.5)|^2 = 9.54$ dB.

To solve the problems of ICI and the off-grid effect, and improve sensing accuracy jointly, in this work, an ICI and off-grid effect suppression algorithm via RDM modeling is proposed. The algorithm provides a fine range and Doppler estimation as well as ICI and off-grid effect suppression. The method and its two variants are introduced in the next section.

III. ICI AND OFF-GRID EFFECT SUPPRESSION ALGORITHM

In this section, the principle of the suppression algorithm is detailed. In addition, its two variants emphasizing the complexity and performance are described.

A. Algorithm with 2D RDM Joint Modeling

The proposed algorithm is target-dependent and achieved by canceling the signal components of detected targets, thus it is deployed after constant false alarm rate (CFAR) detection. The algorithm introduces an iteration, as shown in Fig. 3. Each iteration comprises RDM estimation, CFAR detection, 2D surface fitting & fine estimation, signal reconstruction, and cancellation. The CFAR detection extracts the targets' peak cells, which are the closest to the exact target positions due

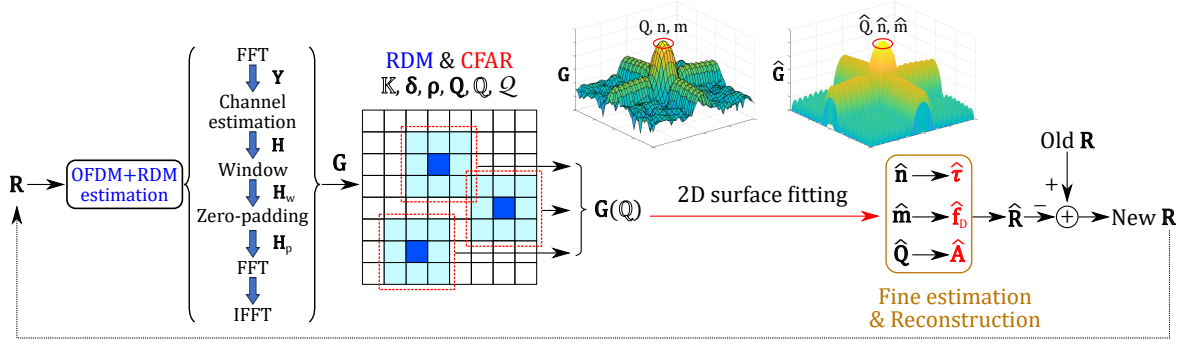


Fig. 3. Simplified flow diagram of the proposed algorithm. The process includes an iteration. In each iteration, the radar receiver performs CFAR detection from the remaining RDM $\mathbf{R}$. The RDM is modeled in surface fitting, where the fine delay and Doppler estimation are implemented simultaneously. The received signal is then reconstructed and canceled from $\mathbf{R}$. The iteration continues until no new target is detected in CFAR.

to the convexity of the window functions, while their adjacent cells may also be detected but can be removed by the peak finder. The on-grid positions on the RDM are given by

$$\boldsymbol{\delta} = [\delta_0, \dots, \delta_{K-1}]^T, \quad (15a)$$

$$\boldsymbol{\rho} = [\rho_0, \dots, \rho_{K-1}]^T, \quad (15b)$$

where K is the number of detected targets in CFAR, collected in the set $\hat{\mathbb{K}}$. For each target k , the peak point and its adjacent eight points are extracted, forming a 3×3 matrix, denoted as $\mathbf{Q}_k$, enclosing the positions of the targets in their respective boundaries, where the center cell $\mathbf{G}(\delta_k, \rho_k)$ is the peak of $\mathbf{G}_k$. The set of cells belonging to $\mathbf{Q}_k$ form a discrete area $\mathbb{Q}_k$, i.e.,

$$\mathbb{Q}_k = \{(n, m) | n \in [\delta_k - 1, \delta_k + 1] \ \& \ m \in [\rho_k - 1, \rho_k + 1], \\ n, m \in \mathbb{Z}\}, \quad (16)$$

and the union of all $\mathbb{Q}_k$ is written as $\mathbb{Q}$:

$$\mathbb{Q} = \bigcup_{k=0}^{K-1} \mathbb{Q}_k. \quad (17)$$

$\mathbb{Q}_k$ is defined as the continuous area enclosed by the boundary of $\mathbb{Q}_k$ and thus $(\delta_k + \varepsilon_k, \rho_k + \epsilon_k) \in \mathbb{Q}_k$, i.e., the solution area of the k -th target:

$$\mathbb{Q}_k = \{(\hat{n}, \hat{m}) | \hat{n} \in [\delta_k - 1, \delta_k + 1] \ \& \ \hat{m} \in [\rho_k - 1, \rho_k + 1], \\ \hat{n}, \hat{m} \in \mathbb{R}\}. \quad (18)$$

It is necessary to ensure that all the components of $\mathbf{Q}_k$ are located in the mainlobe. On the one hand, this forms a convex optimization problem, limiting the estimated values in the mainlobe; on the other hand, the sample power of the mainlobe is significantly higher than that in sidelobes, enhancing the robustness against noise. Therefore, the zero-padding factors p_d and p_v should be at least one. For the consideration of computational effort, $p_d = p_v = 1$ is chosen in this work.

As the ICI is approximately regarded as AWGN and unpredictable without fine estimated signal parameters, the RDM is modeled with the same shape as $\mathbf{G}_{\text{des}}$ in (14):

$$\hat{\mathbf{G}}(n, m, \hat{\mathbf{Q}}, \hat{\mathbf{n}}, \hat{\mathbf{m}}) = \sum_{k=0}^{K-1} \hat{Q}_k \mathbf{W}\left(\frac{n - \hat{n}_k}{1 + p_d}, \frac{m - \hat{m}_k}{1 + p_v}\right), \quad (19)$$

where $\hat{\mathbf{Q}} = [\hat{Q}_0, \dots, \hat{Q}_{K-1}]$, $\hat{\mathbf{n}} = [\hat{n}_0, \dots, \hat{n}_{K-1}]$, and $\hat{\mathbf{m}} = [\hat{m}_0, \dots, \hat{m}_{K-1}]$ are the amplitude and shifts in the range and velocity profiles that need to be optimized. In consequence, the fine estimation problem is transformed into a least square (LS) optimization problem minimizing the difference between $\mathbf{G}$ and $\hat{\mathbf{G}}$, i.e.,

$$(\hat{\mathbf{Q}}, \hat{\mathbf{n}}, \hat{\mathbf{m}}) = \underset{\hat{\mathbf{Q}}, \hat{\mathbf{n}}, \hat{\mathbf{m}}}{\operatorname{argmin}} \|\mathbf{G}(\mathbb{Q}) - \hat{\mathbf{G}}(\mathbb{Q}, \hat{\mathbf{Q}}, \hat{\mathbf{n}}, \hat{\mathbf{m}})\|, \quad (20)$$

$$\text{subject to } (\hat{n}_k, \hat{m}_k) \in \mathbb{Q}_k, \ \forall k \in [0, K-1], \ k \in \mathbb{Z}. \quad (20a)$$

The optimization involves all cells on $\mathbf{G}$ and $\hat{\mathbf{G}}$ with indices $(n, m) \in \mathbb{Q}$, while $\hat{\mathbf{n}}$ and $\hat{\mathbf{m}}$ are limited in the mainlobes. The initial values of $(\hat{\mathbf{Q}}, \hat{\mathbf{n}}, \hat{\mathbf{m}})$ are set to the on-grid ones $(\mathbf{G}(\boldsymbol{\delta}, \boldsymbol{\rho}), \boldsymbol{\delta}, \boldsymbol{\rho})$ in the optimization. This results in a 2D surface fitting and convex optimization problem, which can be efficiently solved by trust-region method [30], Nelder-Mead simplex method [31], or optimization toolbox such as CVX [32], etc. Generally, the result of $(\hat{\mathbf{n}}, \hat{\mathbf{m}})$ are expected to be equal or very close to $(\boldsymbol{\delta} + \varepsilon, \boldsymbol{\rho} + \epsilon)$.

Afterward, the fine-estimated delay (range), Doppler (radial velocity), and CA are calculated by

$$\hat{\tau} = (\hat{\mathbf{n}} - 1)\Delta\tau_p, \quad (21a)$$

$$\hat{f}_D = (\hat{\mathbf{m}} - 1)\Delta f_{D,p} - f_u, \quad (21b)$$

$$\hat{\mathbf{A}} = \frac{1}{NM} \hat{\mathbf{Q}} \odot \operatorname{sinc}\left(-\frac{\hat{f}_D}{\Delta f}\right), \quad (21c)$$

where the term of $-f_u$ in (21b) takes the minus Doppler into account, and the term $\operatorname{sinc}(-\frac{\hat{f}_D}{\Delta f})$ removes the peak loss due to the Doppler intolerance. The division of NM cancels the increased power by processing gain. The set of detected targets in this iteration $\hat{\mathbb{K}}$ is then merged into the set $\mathbb{K}$ collecting all detected targets.

After fine estimation, the signal components of the targets are reconstructed:

$$\hat{\mathbf{R}} = \sum_{k=0}^{K-1} \hat{A}_k \mathbf{D}_N\left(\frac{\hat{f}_{D,k}}{N}\right) \mathbf{F}_N^{-1} \mathbf{D}_N(-\hat{\tau}_k) \cdot \mathbf{X} \mathbf{D}_M(\hat{f}_{D,k} \alpha), \quad (22)$$

where $\hat{f}_{D,k} = \hat{f}_{D,k}/\Delta f$ and $\hat{\tau}_k = \hat{\tau}_k/T$ denote the normalized estimated Doppler shift and delay. At the end of each iteration, $\hat{\mathbf{R}}$ is canceled from the received signal $\mathbf{R}$, i.e.,

$$\mathbf{R} = \mathbf{R} - \hat{\mathbf{R}}. \quad (23)$$

Algorithm 1: ICI and off-grid effect suppression via RDM modeling.

```

1 Initiate Set  $K = 0$ ,  $\mathbb{K} = \emptyset$ , received signal  $\mathbf{R}$ .
2 while True do
3   OFDM demodulation:  $\mathbf{Y} \leftarrow \mathbf{R}$  (6).
4   RDM estimation:  $\mathbf{G} \leftarrow \mathbf{Y}$  (7)-(11), with padding
     factors  $p_d = p_v = 1$ .
5   CFAR target detection  $\rightarrow \hat{\mathbb{K}}, K, \delta, \rho$ .
6   if  $\hat{\mathbb{K}} = \{\emptyset\}$  then
7     | Break
8   end
9   Extract the peak cells and their adjacent cells,
     forming  $\mathbb{Q}$  (16)-(17).
10  Model  $\hat{\mathbf{G}}$  with initial values equal to on-grid ones
     (19).
11  Solve convex optimization problem (20) and extract
      $\hat{\mathbf{Q}}, \hat{\mathbf{n}}, \hat{\mathbf{m}}$ .
12  Fine estimation:  $\hat{\tau}, \hat{f}_D, \hat{\mathbf{A}}$  (21).
13  Update the set of detected targets:  $\mathbb{K} = \mathbb{K} \cup \hat{\mathbb{K}}$ 
14  Signal reconstruction of  $\hat{\mathbf{R}}$  (22).
15  Cancellation: New  $\mathbf{R}$  (23).
16 end

```

The cancellation process removes the components of the detected signals, including their ICI and off-grid effect. In the next iteration, the weaker targets hiding under noise become observable. The iteration is implemented until no new target is found in CFAR detection. The algorithm is summarized in Algorithm 1.

In this paper, $O(\cdot)$ is utilized to count the computational complexity, and the calculations with complexity related to $O(1)$ are ignored. The complexity of OFDM demodulation is $O(NM \log N)$. The RDM estimation includes channel estimation and window function with respective complexity of $O(NM)$, and FFT & IFFT with complexity of $O(N_p M_p \log(N_p M_p))$. The complexity of CFAR detection with a window size of W_C is $O(W_C N_p M_p)$. If $p_d = p_v = 1$, the complexity of conventional OFDM radar receivers is $O(NM(2 + \log N + 4 \log(4NM) + 4W_C)) \triangleq O(NM(\log(N^2 M) + W_C))$. For the algorithm, assuming I iterations are executed in the operation. In the i -th iteration, K_i targets are detected. The complexity of convex optimization using e.g., the Nelder-Mead simplex method is $O(3K_i E) \triangleq O(K_i^2)$, where $E \approx O(3K_i)$ is the iteration time in the optimization using the Nelder-Mead simplex method with $3K_i$ variances and fixed accuracy [33]. Signal reconstruction and cancellation take the complexity of $O(K_i NM)$ for NM samples. Therefore, the algorithm increases the complexity to $O(INM(\log(N^2 M) + W_C) + \sum_{i=1}^I K_i(K_i + NM))$. In particular, in the case of $I = 1$, the complexity is $O(K^2 + NM(\log(N^2 M) + W_C + K))$, where RDM estimation is still the dominant.

B. Simplification: Separated 1D Optimizations

The window functions extremely suppress the sidelobes. For instance, the PSLR of Hamming window is -42.7 dB, thus

the influence between the targets on the RDM is negligible when their mainlobes are not overlapped. Therefore, the joint optimization of K targets' parameters can be divided into K independent optimizations. In this case, the 2D surface fitting can be further simplified into two 1D curve fitting problems for range and velocity profiles. The extracted samples for the k -th target in the range and velocity profiles are given by $\mathbf{q}_{d,k}$ and $\mathbf{q}_{v,k}$, respectively. The corresponding indices in the range and velocity profiles are given by

$$\mathbb{Q}_{d,k} = \{n | n \in [\delta_k - 1, \delta_k + 1], n \in \mathbb{Z}\}, \quad (24a)$$

$$\mathbb{Q}_{v,k} = \{m | m \in [\rho_k - 1, \rho_k + 1], m \in \mathbb{Z}\}. \quad (24b)$$

The optimization limitation areas for the k -th target are written as

$$\mathbb{Q}_{d,k} = \{\hat{n} | \hat{n} \in [\delta_k - 1, \delta_k + 1], \hat{n} \in \mathbb{R}\}, \quad (25a)$$

$$\mathbb{Q}_{v,k} = \{\hat{m} | \hat{m} \in [\rho_k - 1, \rho_k + 1], \hat{m} \in \mathbb{R}\}. \quad (25b)$$

The range and velocity profiles of the k -th target are modeled referring to $\hat{\mathbf{G}}$:

$$\hat{\mathbf{g}}_{d,k}(n, \hat{q}_{d,k}, \hat{n}_k) = \hat{q}_{d,k} \mathbf{w}_d \left(\frac{n - \hat{n}_k}{1 + p_d} \right), \quad (26a)$$

$$\hat{\mathbf{g}}_{v,k}(m, \hat{q}_{v,k}, \hat{m}_k) = \hat{q}_{v,k} \mathbf{w}_v \left(\frac{m - \hat{m}_k}{1 + p_v} \right). \quad (26b)$$

The 2D joint surface fitting problem then is divided into $K \times 2$ independent curve fitting problems:

$$(\hat{q}_{d,k}, \hat{n}_k) = \underset{\hat{q}_{d,k}, \hat{n}_k}{\operatorname{argmin}} \|\mathbf{q}_{d,k} - \hat{\mathbf{g}}_{d,k}(\mathbb{Q}_{d,k}, \hat{q}_{d,k}, \hat{n}_k)\|, \quad (27)$$

$$\text{subject to } \hat{n}_k \in \mathbb{Q}_{d,k}. \quad (27a)$$

$$(\hat{q}_{v,k}, \hat{m}_k) = \underset{\hat{q}_{v,k}, \hat{m}_k}{\operatorname{argmin}} \|\mathbf{q}_{v,k} - \hat{\mathbf{g}}_{v,k}(\mathbb{Q}_{v,k}, \hat{q}_{v,k}, \hat{m}_k)\|, \quad (28)$$

$$\text{subject to } \hat{m}_k \in \mathbb{Q}_{v,k}. \quad (28a)$$

The fine-estimated delay and Doppler are then given by

$$\hat{\tau}_k = (\hat{n}_k - 1) \Delta \tau_p, \quad (29a)$$

$$\hat{f}_{D,k} = (\hat{m}_k - 1) \Delta f_{D,p} - f_u. \quad (29b)$$

The resulting peaks $\hat{q}_{d,k}$ $\hat{q}_{v,k}$ in the range and velocity profiles suffer from the influence of the off-grid Doppler and delay, respectively. Since the off-grid delay and Doppler are independent, CA can be calculated by either compensating the impact of the off-grid range on $\hat{q}_{v,k}$ or the impact of the off-grid Doppler on $\hat{q}_{d,k}$, i.e.,

$$\hat{\mathbf{A}}_k = \frac{\hat{q}_{v,k} \hat{q}_{d,k}}{\mathbf{G}(\delta_k, \rho_k) \cdot NM \operatorname{sinc}(-\frac{\hat{f}_{D,k}}{\Delta f})}. \quad (30)$$

The simplified algorithm is summarized in Algorithm 2. The difference between it and Algorithm 1 exists in the modeling and optimization processes. Afterward, the estimated signals are reconstructed and canceled from the received signal according to (22) and (23).

The complexity of 1D convex optimization using e.g. Nelder-Mead method is $O(2E) \triangleq O(1)$, where $E \approx O(2)$ with fixed accuracy. In the i -th iteration, the optimization for K_i targets leads to a complexity of $O(K_i)$. Consequently, the complexity

Algorithm 2: Simplified algorithm with separated targets and 1D profiles modeling.

```

1 Initiate Set  $K = 0$ ,  $\mathbb{K} = \emptyset$ , received signal  $\mathbf{R}$ .
2 while True do
3   OFDM demodulation:  $\mathbf{Y} \leftarrow \mathbf{R}$  (6).
4   RDM estimation:  $\mathbf{G} \leftarrow \mathbf{Y}$  (7)-(11), with padding
     factors  $p_d = p_v = 1$ .
5   CFAR target detection  $\rightarrow \hat{\mathbb{K}}, K, \delta, \rho$ .
6   if  $\hat{\mathbb{K}} = \{\emptyset\}$  then
7     Break
8   end
9   for  $k = 0 \rightarrow K - 1$  do
10    Extract the peak cells and their adjacent cells,
        forming  $\mathbb{Q}_{d,k}$  and  $\mathbb{Q}_{v,k}$  (24a)-(24b).
11    Model profiles  $\hat{\mathbf{g}}_{d,k}$  and  $\hat{\mathbf{g}}_{v,k}$  with initial values
        equal to on-grid ones (26a)-(26b).
12    Solve 1D convex optimization problems
        (27)-(28) and extract  $\hat{q}_{d,k}, \hat{n}_k, \hat{q}_{v,k}, \hat{m}_k$ .
13    Fine estimation:  $\hat{\tau}_k, \hat{f}_{D,k}, \hat{A}_k$  (29)-(30).
14  end
15  Update the set of detected targets:  $\mathbb{K} = \mathbb{K} \cup \hat{\mathbb{K}}$ .
16  Signal reconstruction of  $\hat{\mathbf{R}}$  (22).
17  Cancellation: New  $\mathbf{R}$  (23).
18 end

```

of the simplified method is $O(INM(\log(N^2M) + W_C) + \sum_{i=1}^I K_i(NM))$. The simplification is relatively computing economical when the number of targets is high.

C. Accuracy Improvement: CA Estimation via RDM Reconstruction

One shortcoming of Algorithms 1 and 2 is that the optimization process is affected by the ICI, potentially increasing error in reconstructed signals. While the impact on delay and Doppler estimation is difficult to suppress since the ICI cannot be directly removed from $\mathbf{G}$, the CA accuracy can be further improved by combining with the ICI suppression method proposed in [20], at the expense of increased complexity.

As $\mathbf{G}$ is a linear combination of the RDMs contributed by the targets plus noise components, the idea is to calculate $\hat{\mathbf{A}}$ by minimizing the difference between $\mathbf{G}$ and the linear combination of the reconstructed RDMs. The adapted algorithm (see Algorithm 3) shares the same process with Algorithm 1 before fine estimation of the signal parameters. After surface fitting (20), the fine-estimated delay and Doppler are calculated by (21), while the CA is not calculated in this step. The signal reconstruction process is implemented for each target separately and only involves the estimated delay and Doppler, resulting in delay-Doppler signals (DDSs). The DDS of the k -th target is given by

$$\bar{\mathbf{R}}_k = \mathbf{D}_N(\frac{\tilde{f}_{D,k}}{N})\mathbf{F}_N^{-1}\mathbf{D}_N(-\tilde{\tau}_k) \cdot \mathbf{X}\mathbf{D}_M(\tilde{f}_{D,k}\alpha). \quad (31)$$

Afterward, the DDSs separately undergo the OFDM demodulation and RDM estimation processes in (6), (7)-(11), resulting

Algorithm 3: Improved algorithm with RDM reconstruction.

```

1 Initiate Set  $K = 0$ ,  $\mathbb{K} = \emptyset$ , received signal  $\mathbf{R}$ .
2 while True do
3   OFDM demodulation:  $\mathbf{Y} \leftarrow \mathbf{R}$  (6).
4   RDM estimation:  $\mathbf{G} \leftarrow \mathbf{Y}$  (7)-(11), with padding
     factors  $p_d = p_v = 1$ .
5   CFAR target detection  $\rightarrow \hat{\mathbb{K}}, K, \delta, \rho$ .
6   if  $\hat{\mathbb{K}} = \{\emptyset\}$  then
7     Break
8   end
9   Form the peaks into an array  $\mathbf{q}$ .
10  Extract the peak cells and their adjacent cells,
     forming  $\mathbb{Q}$  (16)-(17).
11  Model  $\bar{\mathbf{G}}$  with initial values equal to on-grid ones
     (19).
12  Solve convex optimization problem (20) and extract
      $\hat{\mathbf{n}}, \hat{\mathbf{m}}$ .
13  Fine estimation:  $\hat{\tau}, \hat{f}_D$  (21).
14  Build the DDSs  $\bar{\mathbf{R}}_k$  separately (31).
15  OFDM demodulation & RDM estimation:
      $\bar{\mathbf{G}}_k \leftarrow \bar{\mathbf{R}}_k$ .
16  Target cells extraction:  $\mathbf{P} \leftarrow \bar{\mathbf{G}}_k, \forall k = 0, \dots, K - 1$ 
     (32)-(33).
17  CA calculation:  $\hat{\mathbf{A}} \leftarrow \langle \mathbf{P}, \mathbf{q} \rangle$  (35).
18  Update the set of detected targets:  $\mathbb{K} = \mathbb{K} \cup \hat{\mathbb{K}}$ .
19  Signal reconstruction of  $\hat{\mathbf{R}}$  (22).
20  Cancellation: New  $\mathbf{R}$  (23).
21 end

```

in delay-Doppler RDMs (DD-RDMs) $\bar{\mathbf{G}}_k, k = 0, \dots, K - 1$, which include the desired and ICI components.

Next, for each DD-RDM, the cells corresponding to all the K targets are extracted and formed as a 1D array. For instance, the extracted cells of the k -th DD-RDM $\bar{\mathbf{G}}_k$ are formed as

$$\mathbf{p}_k = [\bar{\mathbf{G}}_k(\delta_0, \rho_0), \bar{\mathbf{G}}_k(\delta_1, \rho_1), \dots, \bar{\mathbf{G}}_k(\delta_{K-1}, \rho_{K-1})]^T. \quad (32)$$

The extracted arrays are combined as matrix $\mathbf{P}$:

$$\mathbf{P} = [\mathbf{p}_0, \mathbf{p}_1, \dots, \mathbf{p}_{K-1}]. \quad (33)$$

The on-grid peaks are equal to linear combination of $[\mathbf{p}_0, \dots, \mathbf{p}_{K-1}]$ ignoring additional noise, i.e.,

$$\mathbf{q} = \mathbf{P}\hat{\mathbf{A}}, \quad (34)$$

where $\mathbf{q} = [\mathbf{G}(\delta_0, \rho_0), \mathbf{G}(\delta_1, \rho_1), \dots, \mathbf{G}(\delta_{K-1}, \rho_{K-1})]^T$ denotes the on-grid peaks on $\mathbf{G}$.

Therefore, the calculation of CA results in a linear LS problem solved by the pseudo-inverse of $\mathbf{P}$, i.e.,

$$\hat{\mathbf{A}} = (\mathbf{P}^H\mathbf{P})^{-1}\mathbf{P}^H\mathbf{q}. \quad (35)$$

Finally, the target signal is reconstructed according to (22) and canceled from $\mathbf{R}$ according to (23).

Algorithm 3 introduces additional OFDM demodulation and RDM estimation processes. In the i -th iteration, the reconstruction of DDSs increases the complexity by

TABLE I
OVERVIEW OF THE SYSTEM PARAMETERS.

Parameters	Value
Carrier frequency f_c	77 GHz
Bandwidth B	1 GHz
Number of subcarriers N	2048
Subcarrier distance Δf	488.3 kHz
CP length N_{CP}	512
Symbol duration T	2.048 μ s
Total symbol duration T_O	2.56 μ s
Number of OFDM symbols M	1024
Modulation scheme	QPSK
Window function $\mathbf{W}$	2D Hamming window
IFFT length with padded zeros N_p	4096
FFT length with padded zeros M_p	2048
Maximum unambiguous range r_u	307.0 m
Maximum tolerable range r_{max}	76.7 m
Range resolution Δr	0.150 m
Maximum unambiguous velocity v_u	380.2 m/s
Maximum tolerable velocity v_{max}	95.1 m/s
Velocity resolution Δv	0.743 m/s
Radar processing gain G	60.5 dB

$O(K_i NM)$, while the OFDM demodulation and RDM estimation cost $O(K_i NM \log(N^2 M))$. CA calculation leads to complexity of $O(K_i^3)$. Therefore, the total computational effort is $O(INM(\log(N^2 M) + W_C) + \sum_{i=1}^I K_i(K_i + NM \log(N^2 M) + K_i^2))$, being significantly increased.

IV. NUMERICAL RESULTS

Table I summarizes the OFDM radar parameters configured in this work. The system operates at 77 GHz with a bandwidth of 1 GHz. The number of subcarriers is set to 2048, corresponding to the subcarrier distance of 488.3 kHz. The CP length is set to 512 to avoid the ISI. The radar sensing frame includes 1024 OFDM symbols, and the processing gain of 60.5 dB includes the SNR loss of 2.7 dB due to window function. The 2D Hamming window is applied at the radar receiver. The maximum tolerable range and velocity take the ISI and ICI into consideration, respectively. The ISI can be avoided when the distance is below r_{max} , while reducing velocity can only limit but not eliminate the ICI unless the velocity is reduced to zero. v_{max} can be simply calculated with $f_{D,max} = 0.1 \Delta f$ [34], corresponding to $v_{max} = 95.1$ m/s.

Fig. 4 shows an overview of the improvement provided by Algorithm 1 in a single target scenario. The channel attenuation is normalized to 0 dB. The on-grid and off-grid factors are set to $\delta = 128$, $\rho = 256$, $\varepsilon = \epsilon = 0.5$, corresponding to $\tau \approx 0.25 \tau_{max}$ and $f_D \approx 0.5 f_u$. The channel AWGN is excluded. The sidelobes contributed by the off-grid effects on Fig. 4(a) are obviously higher than other cells where the noise is contributed by the ICI. In contrast, the ICI results in randomly distributed noise on the entire RDM. After canceling the target's contribution, the residual RDM in Fig. 4(b) shows a significant suppression (approx. 50-60 dB) of these two effects, although some traces remain.

The performance of the proposed algorithms in different aspects is evaluated in this section. The investigated performance indicators include behavior in general multi-target scenarios as well as with different Doppler shifts, off-grid effects, and SNR. Note although zero-padding plays a crucial role in

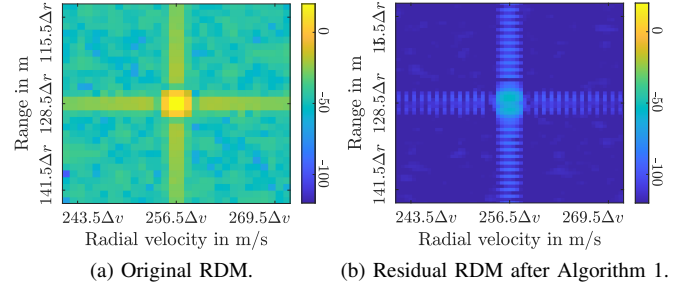


Fig. 4. Single target RDMs of conventional OFDM and Algorithm 1 after cancellation.

proposed algorithms, it leads to increased sidelobes due to increased samples on the transformed window function before the algorithm. For instance, both the red and purple samples in Fig. 2 are extracted in the case of $p_v = 1$, leading to a higher sidelobe level and lower dynamic range than $p_v = 0$. Therefore, the padding factors are set to 0 for the conventional OFDM radar receiver and 1 for the proposed algorithms for the convex optimization process.

A. Behavior in General Multi-Target Scenarios

The general multi-target scenarios are considered in this section, where K targets are generated with randomly and uniformly distributed range ($d \in (0, r_{max}]$), radial velocity ($v \in [-v_u, v_u]$), and phase shift ($\varphi \in [-\pi, \pi]$). The channel attenuation of the reflection paths is normalized to 0 dB to avoid undesired observed fluctuation due to significantly different reception power. The SNR at the receiver side is set to 0 dB. The CFAR detection uses the cell-averaging algorithm with $W_C = 21 \times 21$, and the false alarm rate per cell is set to $1 - 0.9^{\frac{1}{N_p M_p}} = 1.3 \times 10^{-8}$. In the experiment, K is varied from 1 to 10, and the results are obtained from the Monte-Carlo simulation. The dynamic range η is chosen to measure the performance of the algorithms. For the original RDM, η represents the ratio between the minimal target peak and the highest cell power outside the mainlobes. For the improved RDMs, the numerator becomes the lowest value in $\hat{\mathbf{Q}}$ (peaks of the modeled RDM). In addition, the fine-estimation accuracy of delay and Doppler are expressed by normalized mean absolute errors (NMAEs), which is defined by

$$\bar{e}_\tau = \frac{1}{K} \sum_{k=0}^{K-1} \frac{|\hat{\tau}_k - \tau_k|}{\Delta \tau}, \quad (36a)$$

$$\bar{e}_f = \frac{1}{K} \sum_{k=0}^{K-1} \frac{|\hat{f}_{D,k} - f_{D,k}|}{\Delta f_D}. \quad (36b)$$

Fig. 5 illustrates the dynamic range of different systems with increased targets. The increased number of targets leads to reduced dynamic ranges for all the systems. Nevertheless, compared with the original RDM, Algorithms 1 and 2 provide a stable dynamic range gain of $\Delta \eta_1 \approx \Delta \eta_2 > 40$ dB, while the former slightly outperforms the latter, as shown in the zoomed diagram. At the same time, Algorithm 3 provides a further gain of 3.6 dB at $K = 1$ and 4.5 dB in multi-target scenarios than Algorithms 1 and 2.

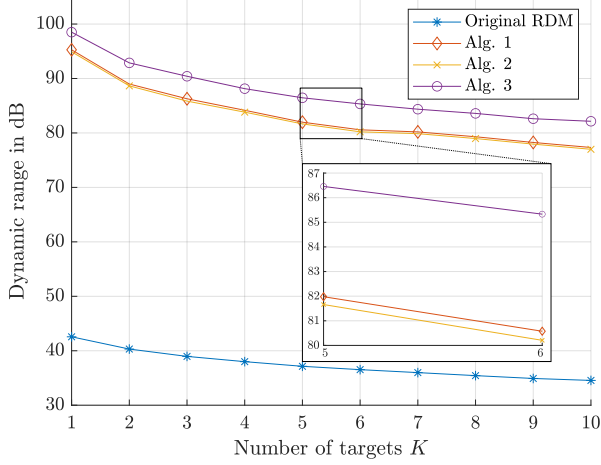


Fig. 5. Dynamic range with the increased number of targets.

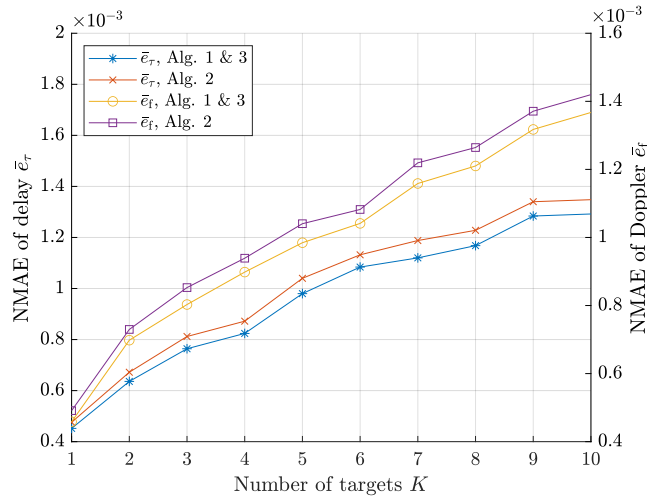

 Fig. 6. Delay and Doppler NMAEs with the increased number of targets. The delay and Doppler NMAEs are measured by the y -axes on the left and right sides, respectively.

Fig. 6 shows the fine-estimation accuracy of delay and Doppler with different numbers of targets. Algorithms 1 & 3 have the same estimation process and thus are drawn by the same curve. The y -axes on the left and right sides represent $\bar{\epsilon}_\tau$ and $\bar{\epsilon}_f$, respectively. The delay and Doppler NMAEs increase with K , but are still under 1.6×10^{-3} even when K reaches 10, showing a reliable performance in multi-target scenarios. Compared with Algorithm 1, the degradation due to the simplification in Algorithm 2 is limited in around 6×10^{-5} and can be ignored.

The trend of performance degradation comes from the growth in ICI due to increased K . While the path gains of the targets are normalized to 0 dB, the noise generated by the ICI increases with K , which may influence the optimization process. However, the degradation is slight and the performance is still safely guaranteed when K grows to 10. The result verifies the applicability in multi-target scenarios.

In the next sections, the simulation with a single target

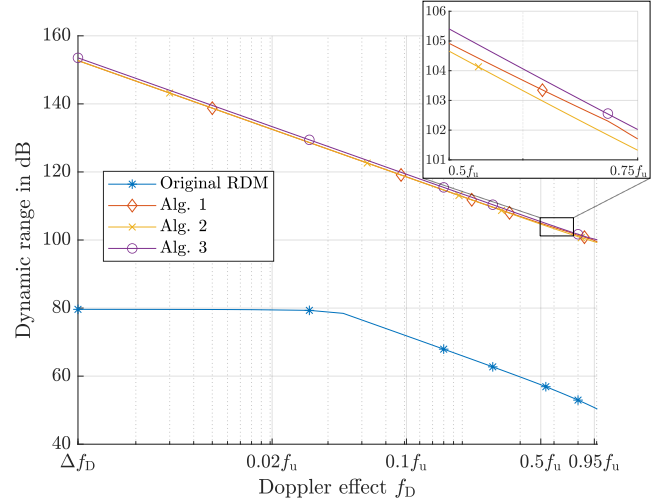


Fig. 7. Dynamic range with the increased Doppler effect.

and performance in specific aspects including ICI/off-grid effect/SNR are implemented and evaluated.

B. Performance of ICI Mitigation

The performance of the proposed algorithms in ICI suppression is evaluated in this section. To intuitively reflect the suppression of ICI, $\epsilon = \epsilon = \sigma_z^2 = 0$ is configured thus the off-grid effect and the channel AWGN cannot influence the result. The distance is set to 19.19 m ($\delta = 128$) to avoid the ISI. The Doppler effect f_D varies from 0 to $0.95 f_u$, where $f_u = \frac{2v_u}{c_0} f_c$ is the maximum unambiguous frequency and c_0 denotes the speed of light in vacuum. The dynamic range η and the NMAEs of delay and Doppler are evaluated.

Fig. 7 describes the relationship between the dynamic range of different systems and the Doppler effect. The dynamic range of original RDM η_0 remains stable at approximately 79 dB when $f_D < 0.02 f_u$. This is because the sidelobe of the Hamming window is the dominant factor limiting the dynamic range when ICI is tiny. As given in Fig. 2, the maximum sidelobe is about -79 dB in the on-grid case, corresponding to η_0 in Fig. 7. With the increased Doppler effect, all four curves show a linearly decreased dynamic range since the samples $\mathbf{G}(\mathbb{Q})$ are influenced by the ICI. However, the proposed algorithms can always increase the dynamic range by at least 48.7 dB compared with the conventional OFDM radar. Even at $f_D = 0.95 f_u$, the dynamic range of the algorithms is still above 100 dB, being approx. 20 dB higher than the maximum achievable dynamic range of the conventional system, showing a reliable performance in ICI suppression. In addition, the three algorithms show an insignificant difference in performance, as shown in the zoomed diagram, implying that the changes in Algorithms 2 and 3 are almost irrelevant to ICI suppression.

The impact of ICI on delay and Doppler fine estimation accuracy is shown in Fig. 8. Clearly, both the delay and Doppler NMAEs grow linearly with the increasing ICI, but cannot exceed 6×10^{-4} even when f_D grows to $0.95 f_u$, showing high robustness against the ICI. Further, it is shown that Algorithm 1 has a slightly higher accuracy than Algorithm 2, but the gap is

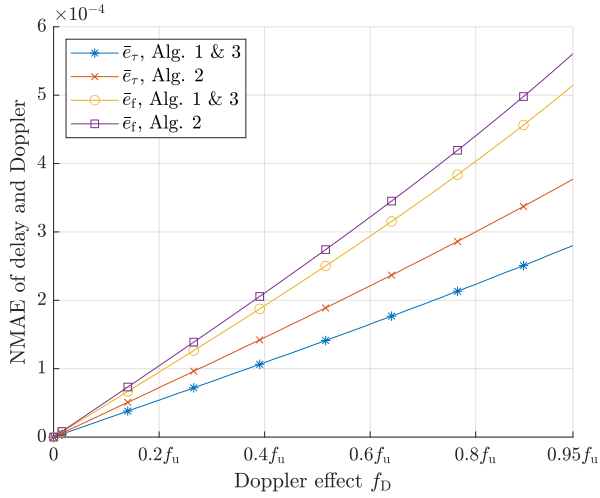


Fig. 8. Delay and Doppler NMAEs with the increased Doppler effect.

less than 10^{-4} in delay and 5×10^{-5} in Doppler. Hence, the difference between the performance of the proposed algorithms in ICI-only environments is negligible.

C. Suppression of the Off-Grid Effect

The performance in off-grid effect suppression is evaluated in this section. The channel AWGN power is set to 0. The on-grid and off-grid factors are calculated based on the conventional OFDM receiver, where $p_d = p_v = 0$. The target distance is set to 19.19 m ($\delta = 128$), and the Doppler shift is set to $0.5f_u$ ($\rho = 256$), since on the one hand, the impact of ICI becomes significant when $f_D = 0.5f_u$, the off-grid compensation performance with the impact of ICI can be observed; on the other hand, the impact of ICI on the dynamic range in the interval of $[f_D - 0.5\Delta f, f_D + 0.5\Delta f]$ is almost unchanged and thus cannot significantly influence the result when f_D is relatively high, as illustrated in Fig. 7. The off-grid factors ε and ϵ are simultaneously grown from -0.5 to 0.5 in the simulation. The dynamic ranges and the NMAEs are recorded.

Fig. 9 shows the dynamic range versus off-grid factors curves. The results with and without algorithms are illustrated in separate diagrams with different scales. The dynamic range of the original RDM stays at around 57.5 dB when $|\varepsilon|$ and $|\epsilon| \leq 0.04$ and shows a slightly decreasing trend in this interval since the ICI is the dominant factor limiting the dynamic range when $|\varepsilon|$ and $|\epsilon|$ are small, and the increased Doppler f_D due to the growth of ϵ slightly influences the dynamic range. When $|\varepsilon|$ and $|\epsilon|$ further rise, the dynamic range is significantly reduced as the increased sidelobes become the dominant noise on the RDM.

On the contrary, the proposed algorithms benefit from extremely increased reliability against the off-grid effect, where the jitter of the dynamic range is limited in 6 dB when ε and ϵ change from -0.5 to 0.5 . Compared with the conventional OFDM receiver, the dynamic range gain $\Delta\eta$ produced by the algorithms grows from approx. 47 dB at $\varepsilon = \epsilon = 0$ to 65 dB at $|\varepsilon| = |\epsilon| = 0.5$, showing a significant improvement. The

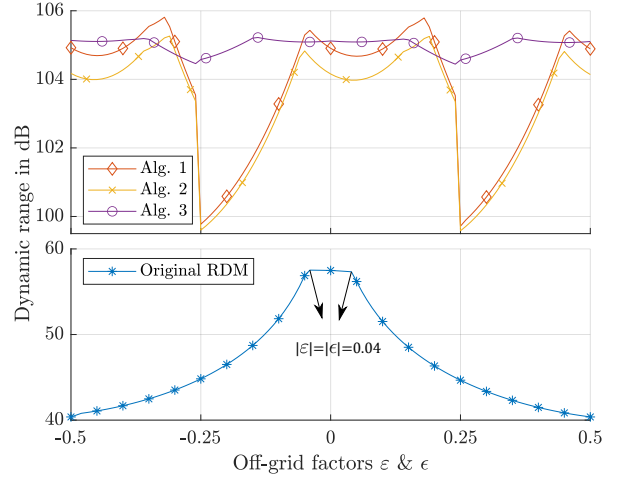


Fig. 9. Dynamic range versus off-grid effect curves. The results with the original RDM and the algorithms are shown in separate diagrams due to the large gap between them and different fluctuation ranges.

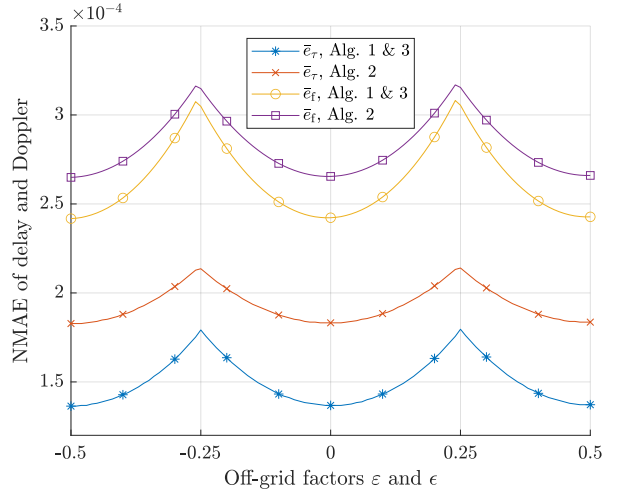


Fig. 10. Delay and Doppler NMAEs with the varied off-grid factors.

dynamic range shows a variation with a periodicity of 0.5, corresponding to the halved grid size due to zero padding. The dynamic ranges of the algorithms reach their lowest at $|\varepsilon| = |\epsilon| = 0.25$, as the off-grid effect is maximized at these points. Algorithms 1 and 2 have a similar performance, while the result of Algorithm 3 is more stable and exempts from the decline in $[-0.29, -0.05]$ and $[0.21, 0.45]$.

The estimation accuracy of delay and Doppler is illustrated in Fig. 10. The jitter of the delay and Doppler NMAEs for each algorithm is less than 7×10^{-5} . The peak of the delay and Doppler NMAEs at $|\varepsilon| = |\epsilon| = 0.25$ are below 2.2×10^{-4} and 3.2×10^{-4} , reflecting an attractive performance in off-grid factor estimation.

D. Impact of Channel AWGN

To evaluate the performance of the proposed algorithms in noisy environments, in this section, the SNR is chosen as the variable. Again, δ and velocity ρ are set to 128 and 256,

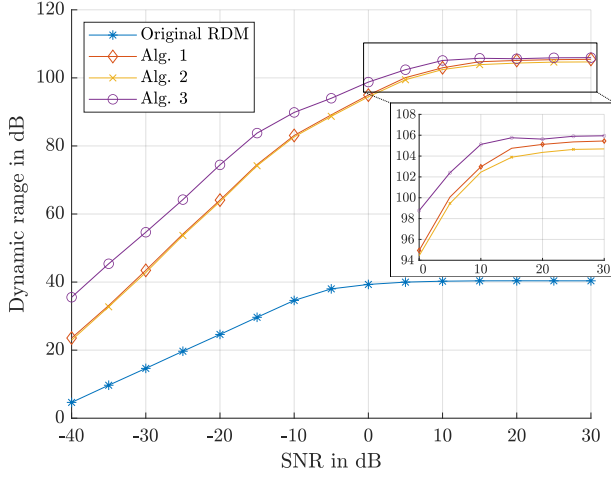


Fig. 11. Impact of SNR on the dynamic ranges.

respectively, and the off-grid components ε and ϵ are set to 0.5. The SNR in the received signal is varied from -40 dB to 30 dB, and the corresponding SNR of the RDM with processing gain falls in the interval of $[20.5, 90.5]$ dB. Fig. 11 shows the relationship between channel SNR and the resulting dynamic ranges. Clearly, both the conventional OFDM receiver and the receiver with the proposed algorithms are impacted by the AWGN, while the latter shows a faster descent when SNR decreases. However, Algorithm 1 and 2 can provide a dynamic range gain of 18.3 dB than the conventional system even when SNR drops to -40 dB, while Algorithm 3 further provides more than 10 dB dynamic range gain when $\text{SNR} < -15$ dB, and the gain grows to 12.6 dB when $\text{SNR} = -40$ dB, reflecting a better applicability in noisy environments.

The dynamic range of the conventional OFDM receiver levels off when SNR reaches 0 dB due to the ICI and the off-grid effect, the receiver with the proposed algorithms increases faster and the upward momentum stops until SNR exceeds 15 dB, where the dynamic range gain provided by the three algorithms are converged to approx. 65 dB, as given in the zoomed diagram, showing a similar performance in noiseless environments.

The delay and Doppler NMAEs are shown in Fig. 12, and their respective y -axes are separated and illustrated on the left and right sides to improve visibility. Although the delay and Doppler NMAEs increase faster when SNR decreases, they are limited below 0.035 even when SNR drops to -40 dB, indicating that the proposed algorithms can provide high accuracy delay and Doppler estimation in noisy environments. In addition, the simplification by separated estimation only leads to a slight accuracy degradation of both the delay and Doppler, as illustrated in the zoomed diagram, i.e., the robustness against noise is not significantly influenced.

E. Separation of Adjacent Targets

Simulation results in Sections IV-A to IV-D indicate that the proposed algorithms can provide significant improvements in dynamic range and estimation accuracy. Algorithm 3

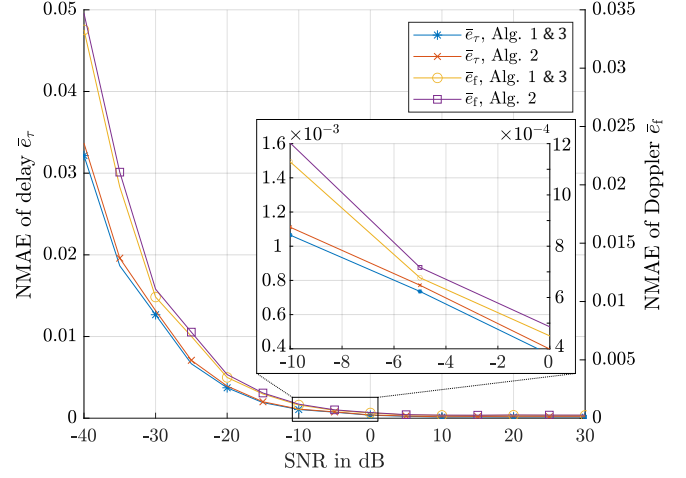


Fig. 12. Delay and Doppler NMAEs versus SNR curves. The difference between the results of Algorithms 1 and 2 is so tiny that they can only be distinguished in a zoomed diagram.

outperforms other algorithms in highly noisy and multi-target scenarios, while Algorithms 1 and 2 show a negligible difference in performance. However, in some realistic use cases, the target points may be located in adjacent cells on the RDM, leading to overlapped mainlobes and strong interference between the targets' contribution cells. Algorithms 1 and 3 take this effect into account, while Algorithm 2 is under the assumption that the target peak cells are located at a sufficient distance from each other so that their mainlobes are not overlapped.

In this section, the impact of the interference between the adjacent targets is evaluated. Two targets with the same channel attenuation (normalized to 0 dB) and Doppler shift ($f_D = 0.5f_u$) and randomly generated phase shifts are set in the simulation. Their delays are set to $128\Delta\tau$ and $(128 + \xi)\Delta\tau$, respectively, where $\xi \in \mathbb{R}$ denotes the separation of the targets in the range domain. The channel AWGN is excluded. In the simulation, the results with $\xi \geq 1$ are investigated, since this guarantees that the target peaks are distinguishable in CFAR detection when $p_d = p_v = 1$. If $\xi < 1$, the peaks may overlap or be adjacent, the algorithm only considers a single target.

The curves of dynamic range versus ξ are shown in Fig. 13. The result of the conventional OFDM radar shows a periodic fluctuation and reaches peaks and troughs when $\xi = n \in \mathbb{Z}$ and $n + 0.5$, respectively, due to the off-grid delay of the second target. The dynamic ranges of Algorithms 1 and 3 show an increasing trend respectively from 93.7 dB and 99.9 dB at $\xi = 1$ to around 98.6 dB and 102.9 dB at $\xi = 1.5$. Afterward, their dynamic ranges show a small irregular fluctuation. However, the result of Algorithm 2 reflects a significant degradation than Algorithms 1 and 3 since the interference between the targets is not considered. The resulting dynamic range is even lower than the original RDM when $\xi < 2$. With the increasing distance between the target cells, the dynamic range increases, and finally shows a periodic fluctuation with an insignificant increasing trend among the periods. Nevertheless, except for $\xi = 2.5$, the peaks still have approx. 10 dB gaps

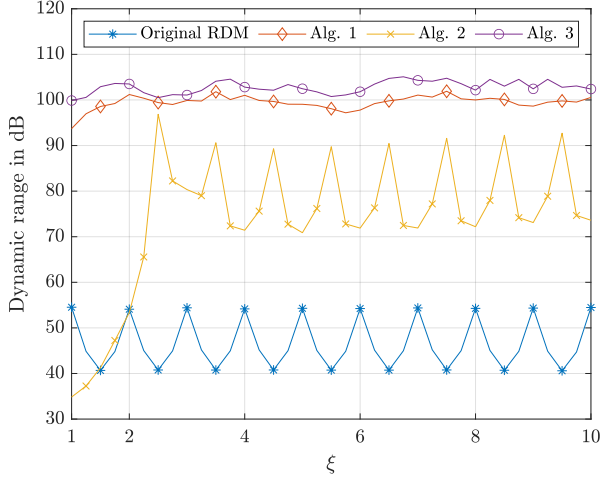


Fig. 13. Impact of adjacent target interference on dynamic range. The two targets lie on the same range profile with varying ξ .

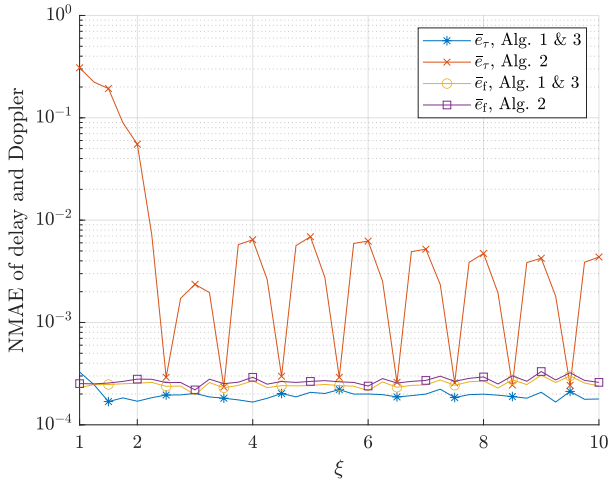


Fig. 14. Impact of the adjacent interference on fine delay and Doppler estimation. The two targets lie on the same range profile with varying distance ξ . The y -axes coordinates are represented in logarithmic form to observe the similarity in shapes of the NMAEs and the window function.

from Algorithm 1, but the difference slightly reduces with more separation between the targets.

The impact of adjacent targets on delay and Doppler estimation is illustrated in Fig. 14. It is shown that the Doppler estimation accuracy is irrelevant to ξ as the targets are located in the same range profile instead of the velocity profile. On the contrary, the interference in the delay domain changes the delay accuracy, leading to degraded dynamic range in Fig. 13. The delay NMAE of Algorithm 1 is only influenced when $\xi < 1.5$ and is still limited in 3.3×10^{-4} when ξ drops to 1, slightly influenced by the adjacent interference. In contrast, Algorithm 2 results in a significantly varying delay error. The delay NMAE of Algorithm 2 raises to 0.308 at $\xi = 1$. When $\xi \geq 2.25$, the delay NMAE shows a periodical fluctuation with peaks and troughs in positions opposite to the dynamic range.

The targets located on the same range profile interfere with each other in delay estimation, and the conclusion can also be

extended to the velocity profile and Doppler estimation. The delay NMAE and dynamic range of Algorithm 2 respectively show a Hamming window-like shape with a shift of 0.5 on the x -axis and an upside-down flipped shape of the former. The reason comes from the interference between the targets. When $\xi < 2.5$, the mainlobe of one target on the RDM is overlapped with at least part of the cells of another target used for curve/surface fitting (i.e., $\mathbf{Q}_k$ or $\mathbf{q}_{d,k}$), the interference thus leads to more significant influence on the fitting process. When $\xi > 2.5$, the optimization-related cells are overlapped with the sidelobes of another target. The center element of $\mathbf{q}_{d,k}$ is stronger against interference than the other two cells due to the highest power. When $\xi = n \in \mathbb{Z}$, the sidelobe trough and peaks of another target are overlapped with the center and its side elements of $\mathbf{q}_{d,k}$, respectively, the impact of interference is maximized. On the contrary, the situation is reversed when $\xi = n + 0.5$, thus the impact of interference reaches the bottom. This results in a dynamic range and delay NMAE with similar shape as the window function with a shift of 0.5 on the x -axis.

V. DISCUSSION

Simulation results show the excellent performance of the proposed algorithms in ICI and off-grid suppression and range/velocity estimation. Algorithm 1 overcomes the difficulties in communications capability [10], [13]–[18], multi-target compatibility [16]–[19], and complexity limitations [20], [21]. In addition, its two variants improved in the respective aspects of complexity and accuracy are introduced. The performance of the algorithms in various scenarios is evaluated in the simulation. The advantages (+) and shortcomings (–) of Algorithm 1 are summarized in the following:

- + Extremely improved accuracy of delay and Doppler.
- + Significantly suppressed ICI and off-grid effect.
- + Only implemented at the radar receiver, thus no influence on communications functionality.
- + Reliable performance in multi-target scenarios.
- + Capability of providing improvement in noisy environments.
- ± Complexity: the complexity depends on the number of iterations I . However, the RDM modeling and fitting process itself only leads to slightly increased complexity. In the case of small I , the increase in complexity is slight.
- Performance depends on target detection. Slight degradation in performance when target peaks are closed to each other ($|\xi| < 1.5$).

The improvements and side-effects of the variants compared with Algorithm 1 are listed as follows:

Algorithm 2:

- + Simplified optimization process and less complexity with slightly impacted performance.
- Significantly degraded performance when target cells are close to each other or in the same range/velocity profile.

Algorithm 3:

- + Significantly increased robustness against noise.
- + Improved performance in multi-target scenarios.
- Extremely increased computational effort.

Considering the communications compatibility and performance in multi-target scenarios, ISAC in traffic applications [6] such as traffic monitoring and automotive radar sensing could be potential use cases for the proposed algorithms, where the number of targets (vehicles, pedestrians, etc.) is generally high, and their velocities are various, requiring a reliable radar receiver with fast reaction. At the same time, the road information system demands excellent communication functionality, which is compatible with the proposed algorithms. However, large objects such as ground and buildings are unexpected in detection and may degrade the algorithm performance since they may contribute to a large number of target points on the RDM with distance $\xi < 1.5$. Thus, clutter [35] detection and removal methods are expected to be implemented in advance.

Another essential application scenario is in unmanned aerial vehicle (UAV)-relevant ISAC systems [8], where the density of targets is generally low and the Doppler shift is various. The small size of UAVs avoids adjacent target interference, thus the proposed methods, especially for Algorithm 2, can be safely applied.

This work does not address the Doppler ambiguity issues since the maximum unambiguous velocity v_u is generally much higher than the maximum speed that vehicles and drones can reach. For instance, v_u in this work is set to 380.2 m/s, while the speed limit for cars is less than or equal to 38.9 m/s (140 km/h) in most countries of the world. Modern automotive radar systems generally operate at 77 GHz with bandwidth from 600 MHz to 4 GHz [2], [3], the corresponding v_u changes from 228.1 m/s to 1520.9 m/s with the number of subcarriers of 2048 and CP length of 512. For 5G network-based downlink sensing, v_u varies from 175 m/s (numerology 0) to 2800.8 m/s (numerology 4) at e.g. 6 GHz in FR1, and from 37.5 m/s (numerology 0), 75 m/s (numerology 1) to 600.2 (numerology 4) at e.g. 28 GHz in FR2 [36], [37]. Most of these systems can safely provide a velocity-unambiguous service in traffic- and vehicular-relevant applications.

VI. CONCLUSION

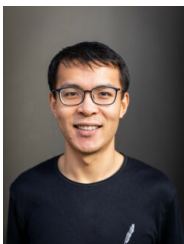
This paper proposes a novel range-Doppler map (RDM) modeling-based algorithm that can suppress the inter-carrier interference (ICI) and the off-grid effect and achieve a super-resolution delay and Doppler estimation for ISAC-enabled OFDM radar. Afterward, its two variants with improvements in complexity or accuracy at the expense of another property are introduced. Simulation results show the proposed algorithms can significantly suppress the ICI and the off-grid effect in multi-target scenarios without impacting communications capability, while the range and velocity errors are reduced to the order of 10^{-3} or less of the resolution. In addition, the use cases such as traffic monitoring and vehicular applications are analyzed. It is explained that the simplified variant Algorithm 2 is expected to provide reliable services in low-density use cases such as unmanned aerial vehicle (UAV) detection, while Algorithm 3 is more robust against noise and ICI. As the proposed algorithms are deployed after target detection, they are modulation-independent and are compatible with

other waveforms such as FMCW and PMCW with minor modifications according to their RDM estimation schemes.

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